

A Finite-Coherence Counterexample to a Global Pressure Claim

Automatic Math Discovery Smoke Test

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Source statement

This packet concerns arXiv:2509.13121, *On the Fixed Point Property in Reflexive Banach Spaces*, by Faruk Alpay and Hamdi Alakkad. In Section 4.4, item 2, the paper discusses finite-dimensional certificates for the diametral pressure functional. The statement spans a page break in the PDF; the stitched screenshot below shows the relevant passage at full page width.

1. **Equivalent reformulation of the gap in Section 3.2.** The obstruction identified in Section 3.2—the failure to control diametral sequences—is precisely the failure of $\mathbf{P} > 0$. Establishing $\mathbf{P}(C_1, x_\infty) > 0$ for all fixed-point-free pairs (C, T) would settle Conjecture 2.11 (p. 5) (Numbering consolidated: Conjecture 2.11 is the main reflexive $\Rightarrow$ FPP conjecture stated in §2.6). Conversely, a counterexample must produce (C, T) with $\mathbf{P} = 0$.
2. **Finite-dimensional certificates.** For each k there exists a k -tuple $(y_i) \subset C_1$ with

$$\inf_{\|a\|_1=1} \left\| \sum_{i=1}^k a_i \frac{y_i - x_\infty}{\Delta} \right\| \geq \theta.$$

This means $\Phi_k(C_1, x_\infty) \geq \theta$. If there exists $\theta > 0$ such that for every k one can find such a k -tuple, then $\mathbf{P}(C_1, x_\infty) = \inf_{k \geq 1} \Phi_k(C_1, x_\infty) \geq \theta$, and by Theorem 4.8 the

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map T must have a fixed point. Thus a positive certificate at each level k yields a verifiable finite-tuple condition guaranteeing the fixed point property. For instance, Proposition 4.16 and Corollary 4.9 show that in any Hilbert space a finite family of unit vectors with sufficiently small mutual coherence admits such certificates: for m vectors with mutual coherence $\mu < 1/(m-1)$ one has a uniform lower bound on $\Phi_k(C_1, x_\infty)$ for $k \leq m$, forcing $\mathbf{P}(C_1, x_\infty) > 0$.

3. **Relation to normal structure.** Normal structure forbids complete diametrality, but it is qualitative. The functional $\mathbf{P}$ quantifies a uniform anti-collapse of signed averages; $\mathbf{P} > 0$ is strictly stronger than normal structure and is tailored to nonexpansive dynamics on C_1 .

FIGURE 1. A screenshot of the relevant passage from the PDF, stitched together.

The specific sentence tested here is the assertion that a finite family of unit vectors in a Hilbert space with sufficiently small mutual coherence gives finite-level certificates which force

$$\mathbf{P}(C_1, x_\infty) > 0.$$

This packet does not address the main reflexive fixed point property conjecture in the paper. It only refutes this finite-certificate-to-global-pressure implication.

Definitions

Following Definition 4.1 of the source paper, for a bounded set C_1 , a point x_∞ , and $\Delta = \text{diam}(C_1)$, set

$$\Phi_k(C_1, x_\infty) = \sup_{y_1, \dots, y_k \in C_1} \inf_{\|a\|_1=1} \left\| \sum_{i=1}^k a_i \frac{y_i - x_\infty}{\Delta} \right\|.$$

The global pressure is

$$\mathbf{P}(C_1, x_\infty) = \inf_{k \geq 1} \Phi_k(C_1, x_\infty).$$

Counterexample

Theorem 1. *Finite low-coherence Hilbert certificates through $k \leq m$ do not force positive global pressure.*

Proof. Let $H = \mathbb{R}^2$ with its Euclidean norm, let $x_\infty = 0$, and let

$$C_1 = \{e_1, e_2\}.$$

The vectors e_1, e_2 are unit vectors and their mutual coherence is

$$\mu = |\langle e_1, e_2 \rangle| = 0 < 1 = \frac{1}{2-1}.$$

Also $\Delta = \text{diam}(C_1) = \|e_1 - e_2\| = \sqrt{2}$.

For $k = 2$, choosing the tuple (e_1, e_2) gives

$$\inf_{|a_1|+|a_2|=1} \left\| \frac{a_1 e_1 + a_2 e_2}{\sqrt{2}} \right\| = \frac{1}{\sqrt{2}} \inf_{|a_1|+|a_2|=1} (a_1^2 + a_2^2)^{1/2} = \frac{1}{2}.$$

Thus the advertised finite-level phenomenon is present: $\Phi_2(C_1, 0) \geq 1/2$, and of course $\Phi_1(C_1, 0) = 1/\sqrt{2}$.

However, for $k = 3$, every 3-tuple chosen from the two-point set C_1 repeats a point. If $y_p = y_q$, choose coefficients

$$a_p = \frac{1}{2}, \quad a_q = -\frac{1}{2}, \quad a_i = 0 \quad (i \neq p, q).$$

Then $\|a\|_1 = 1$ and

$$\sum_{i=1}^3 a_i \frac{y_i - x_\infty}{\Delta} = 0.$$

Hence the inner infimum is zero for every 3-tuple, so $\Phi_3(C_1, 0) = 0$. Consequently

$$\mathbf{P}(C_1, 0) = \inf_{k \geq 1} \Phi_k(C_1, 0) = 0.$$

Therefore the finite low-coherence lower bound for $k \leq m$ does not force positive global pressure. $\square$

Remark 1. *The obstruction is not special to two points. If C_1 is finite with $|C_1| = m$, then every k -tuple with $k > m$ repeats a point, and the same cancellation gives $\Phi_k(C_1, x_\infty) = 0$. This is also consistent with Lemma 2.5 of the source paper. The error is the passage from finite-level certificates to a uniform lower bound over all k .*

Verification code

The packet includes a small exact verifier:

`code/verify_pressure.py`.

It enumerates tuples from $C_1 = \{e_1, e_2\}$ and computes the exact values

$$\Phi_1 = \frac{1}{\sqrt{2}}, \quad \Phi_2 = \frac{1}{2}, \quad \Phi_k = 0 \quad (k \geq 3).$$

Run it with:

```
conda run --no-capture-output -n sandbox python \
    code/verify_pressure.py
```

Novelty and limitation

This is a counterexample to a stated implication in the paper’s programmatic discussion. It should not be counted as a solution to the author-labeled open problems in Section 4.6, and it does not disprove the main reflexive fixed point property conjecture.